Assignment 1 - Git & GitHub Fundamentals

Session 1 - Group D | Individual Lab Report

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Repository: https://github.com/BEH-77/git-session1-Badreddine-El-Habra

1. Repository Setup

I started with the files provided for the assignment and initialized a local Git repository using **git init**. I created a **.gitignore** to exclude **local.env** and ordinary ***.log** files, while keeping **important.log** trackable using **!important.log**.

2. README & Commit History

I wrote a README with my name, GitHub username, a short description of the starter website, the technologies used, a code block, a link, and the provided project image. I organized the work into separate commits for meaningful changes instead of saving everything in a single commit. The complete commit history can be checked in the repository.

3. Selective Staging

I practiced staging only selected changes with **git add <filename>** rather than automatically including every modified file. This lets me commit one tracked file while leaving another modification in the working directory. I also compared an unstaged change using **git diff** and the staged version using **git diff --staged**.

4. Branch & Merge

I worked on the About feature in the **about_feature** branch. I added **about.html**, updated the navigation in **index.html**, and committed the feature. Then I switched back to **main** and merged it using **git merge about_feature**. The terminal output shows a successful fast-forward merge.

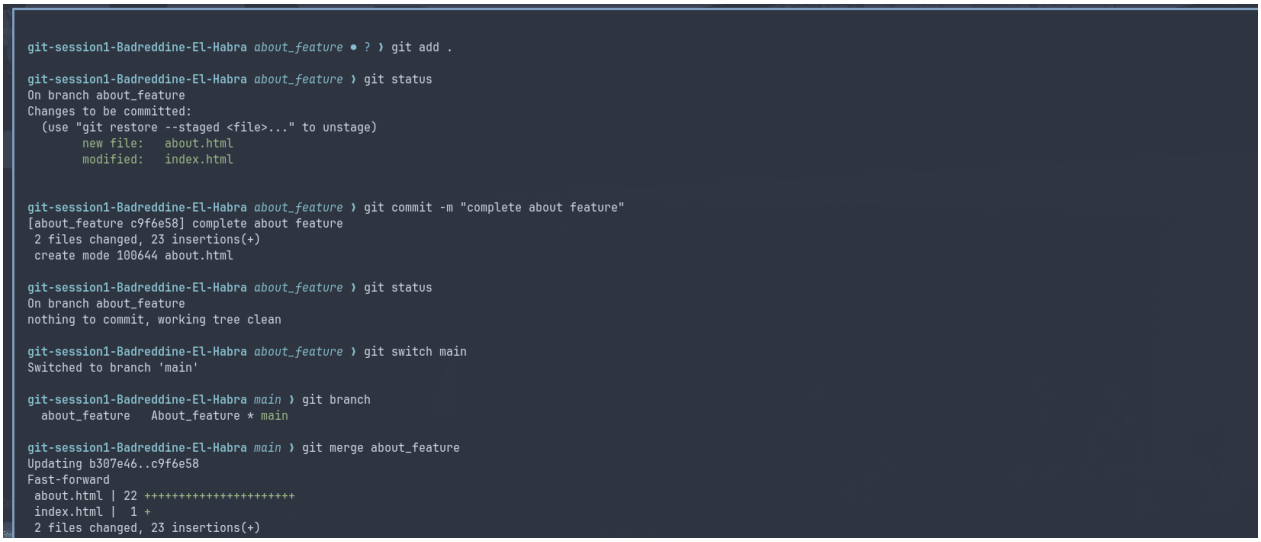
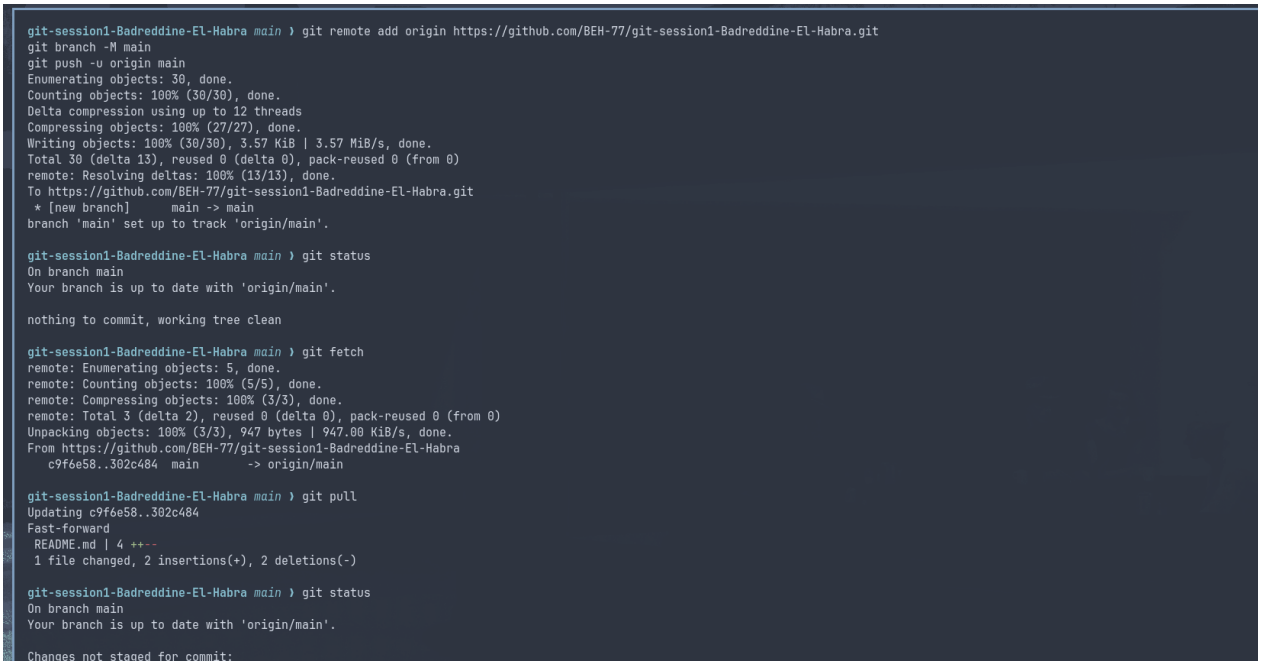


Figure 1. About feature committed on its branch and successfully merged into main.

5. Remote Changes

A **commit** records staged changes in the local repository, while a **push** uploads local commits to GitHub. A **fetch** updates information about remote commits without changing working files, whereas a **pull** brings remote changes into the current branch. I published main, edited the README on GitHub, fetched the new remote commit, and then pulled it locally. The pull output confirms the README was updated.



```
git-session1-Badreddine-El-Habra main ❯ git remote add origin https://github.com/BEH-77/git-session1-Badreddine-El-Habra.git
git branch -M main
git push -u origin main
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 12 threads
Compressing objects: 100% (27/27), done.
Writing objects: 100% (30/30), 3.57 KiB | 3.57 MiB/s, done.
Total 30 (delta 13), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (13/13), done.
To https://github.com/BEH-77/git-session1-Badreddine-El-Habra.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

git-session1-Badreddine-El-Habra main ❯ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

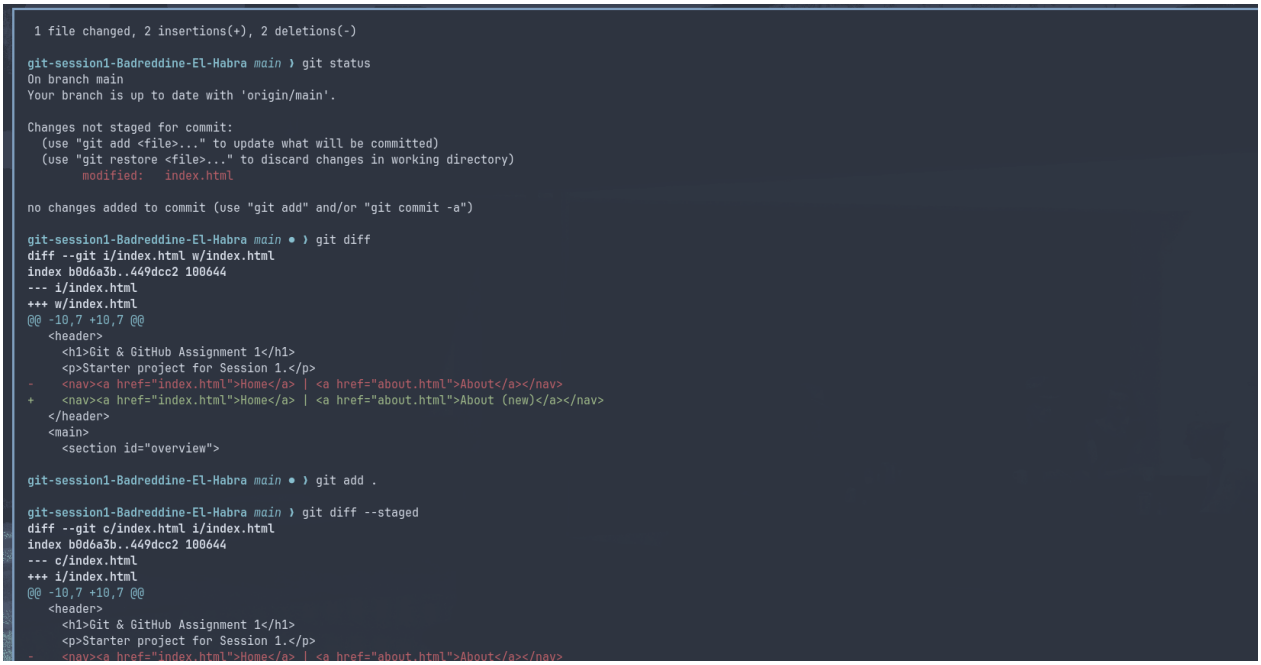
git-session1-Badreddine-El-Habra main ❯ git fetch
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 947 bytes | 947.00 KiB/s, done.
From https://github.com/BEH-77/git-session1-Badreddine-El-Habra
   c9f6e58..302c484  main       -> origin/main

git-session1-Badreddine-El-Habra main ❯ git pull
Updating c9f6e58..302c484
Fast-forward
 README.md | 4 ++--
 1 file changed, 2 insertions(+), 2 deletions(-)

git-session1-Badreddine-El-Habra main ❯ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
```

Figure 2. Initial push, fetch of a newer remote commit, and pull updating README.md.



```
 1 file changed, 2 insertions(+), 2 deletions(-)

git-session1-Badreddine-El-Habra main ❯ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")

git-session1-Badreddine-El-Habra main ● ❯ git diff
diff --git i/index.html w/index.html
index b0d6a3b..449dcc2 100644
--- i/index.html
+++ w/index.html
@@ -10,7 +10,7 @@
   <header>
     <h1>Git & GitHub Assignment 1</h1>
     <p>Starter project for Session 1.</p>
-    <nav><a href="index.html">Home</a> | <a href="about.html">About</a></nav>
+    <nav><a href="index.html">Home</a> | <a href="about.html">About (new)</a></nav>
   </header>
   <main>
     <section id="overview">

git-session1-Badreddine-El-Habra main ● ❯ git add .

git-session1-Badreddine-El-Habra main ❯ git diff --staged
diff --git c/index.html i/index.html
index b0d6a3b..449dcc2 100644
--- c/index.html
+++ i/index.html
@@ -10,7 +10,7 @@
   <header>
     <h1>Git & GitHub Assignment 1</h1>
     <p>Starter project for Session 1.</p>
-    <nav><a href="index.html">Home</a> | <a href="about.html">About</a></nav>
```

Figure 3. Unstaged and staged diffs for index.html, before and after git add.

6. Fair Use of AI

I used AI to help generate the content for the About page and to help write and organize this report. My questions focused on creating a simple About page that fits the starter website and explaining the Git steps clearly in the report template. I used the suggestions as a starting point, reviewed the wording, and checked the Git explanations against the commands I ran and their terminal output. I was responsible for carrying out the Git operations and checking the results myself.

7. Sources & Reflection

Sources: Session 1 Git/GitHub slides, Assignment 1 instructions and report template, Git documentation (git-scm.com/docs), and my terminal output. The trickiest part for me was understanding the difference between changes in the working directory, staged changes, and commits. If Git behaves unexpectedly, I would first run **git status**, check the current branch, and inspect the relevant diff before making another change.